Sets & Relation: Formula Sheet

1. Sets and subsets

Subsets: 2^n; proper / non-empty: $2^n - 1$
Power set: $n(P(A)) = 2^n,\ n(P(P(A))) = 2^{2^n}$
Containing k given elements: 2^{n-k}
Ordered pairs (Y, Z), $Y \cap Z = \phi$: 3^n

2. Operations

Difference: $A - B = A \cap B'$
Symmetric difference: $A \,\Delta\, B = (A - B) \cup (B - A)$
De Morgan:

$$(A \cup B)' = A' \cap B',\ (A \cap B)' = A' \cup B'$$

Distributive:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributive:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

3. Number of elements

Two sets: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Only A: $n(A - B) = n(A) - n(A \cap B)$

$$n(A \,\Delta\, B) = n(A) + n(B) - 2n(A \cap B)$$

None: $n(A' \cap B') = n(U) - n(A \cup B)$
Three sets:

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

Exactly two:

$$\sum n(A \cap B) - 3n(A \cap B \cap C)$$

Exactly one:

$$\sum n(A) - 2\sum n(A \cap B) + 3n(A \cap B \cap C)$$

4. Bounds and multiples

Intersection:

$$\max(0, n(A)+n(B)-n(U)) \le n(A \cap B) \le \min(n(A), n(B))$$

Union:

$$\max(n(A), n(B)) \le n(A \cup B) \le \min(n(A) + n(B), n(U))$$

Multiples of k in $1..N$: $[N/k]$

$$N_a \cap N_b = N_{\mathrm{lcm}(a,b)}$$

5. Cartesian product

$$n(A \times B) = n(A)\, n(B)$$

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

Common to $A \times B$, $B \times A$: $n(A \cap B)^2$

6. Relations

Relations $A \to B$: 2^{mn}; on A: 2^{n^2}
Reflexive: $2^{n(n-1)}$
Symmetric: $2^{n(n+1)/2}$
Reflexive and symmetric: $2^{n(n-1)/2}$
Anti-symmetric: $2^n\, 3^{n(n-1)/2}$
Equivalence: B_n; $B_3 = 5,\ B_4 = 15$
Inverse: $R^{-1} = \{(b, a) : (a, b) \in R\}$
Equivalence class: $[a] = \{x : xRa\}$